ICAR-National Institute of Agricultural Economics and Policy Research (NIAP)

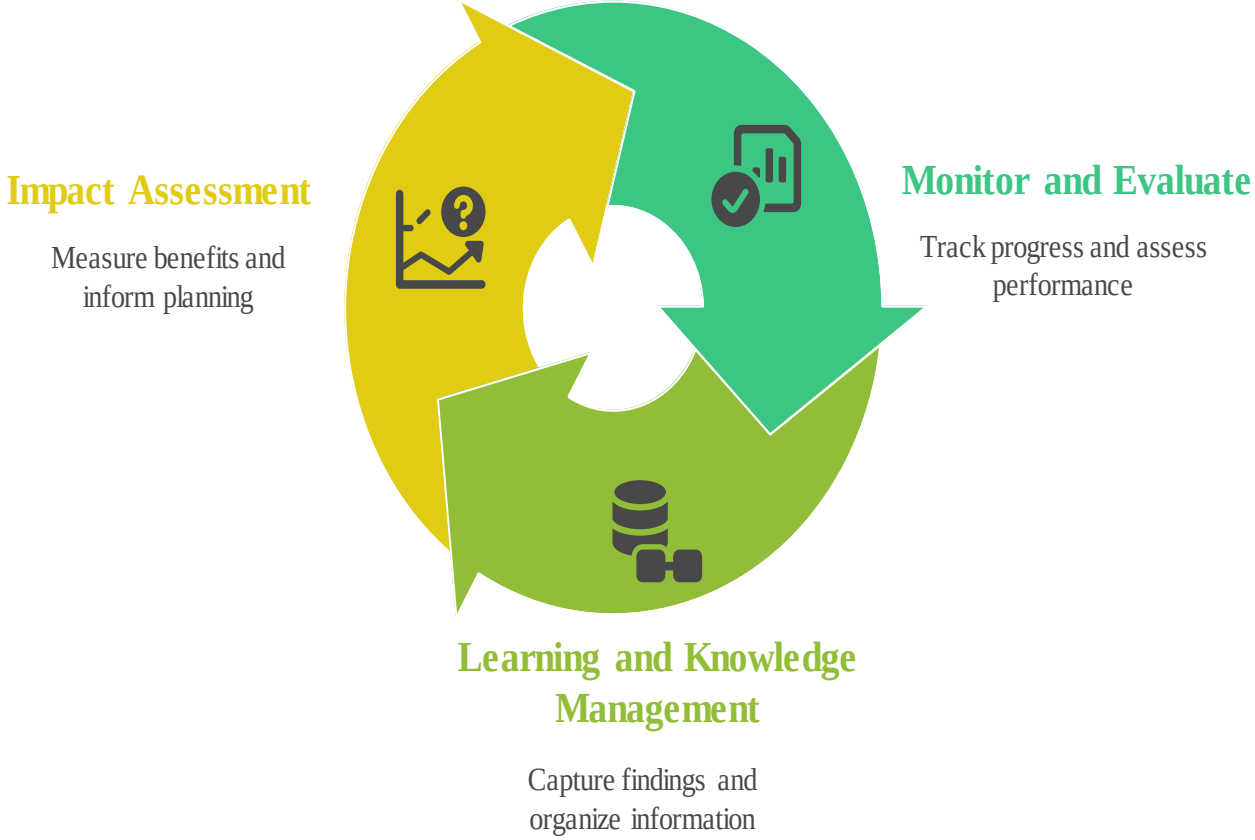
Monitoring, Evaluation, Learning and Impact Assessment (MELIA) unit

Concept note

27-04-2026

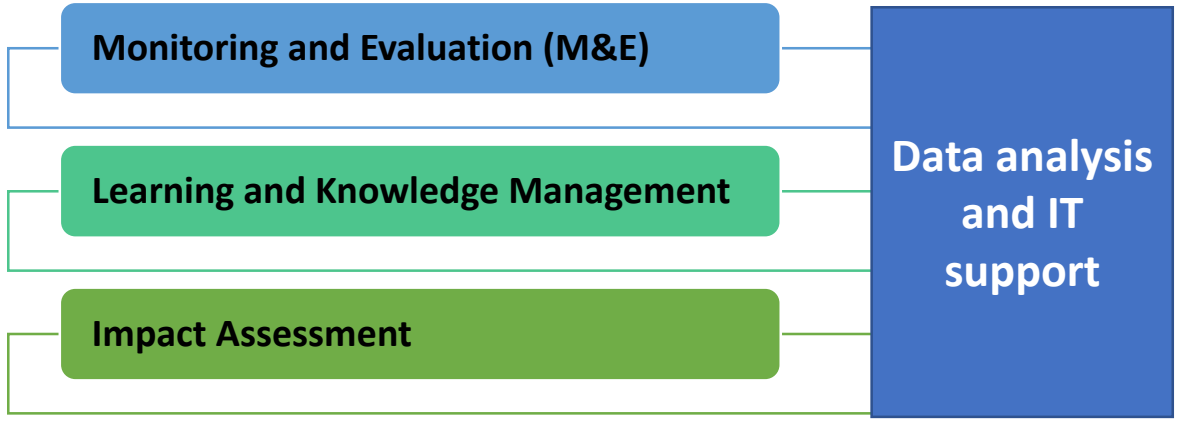
Outline

- Purpose: Introduction to MELIA
- Function: Framework
- Set-up: Operationalizing
- How its run: Structure of MELIA
- Expectation: Advantages/timelines



Purpose: Introduction to MELIA

- Monitoring, Evaluation, Learning and Impact Assessment (MELIA) unit
- **monitor** project implementation against set targets, timelines and resources, evaluate the effectiveness, efficiency, relevance and sustainability of the projects
- **learn** from the success and challenges to inform adaptive management and improvement
- access **impact** of the ICAR technologies and initiatives



Monitoring and Evaluation (M&E)

- Projects and programmes under the ICAR would be systematically tracked and assessed based on key performance indicators (KPIs)
- KPIs are **S**pecific, **M**easurable, **A**chievable, **R**elevant and **T**ime-bound metrics used to track the progress, performance, and result of a project/programme against its objectives
- Monitoring would help research managers and principal investigators of the projects to track progress and to identify whether the resources are being used efficiently, whether activities on schedule and outputs are delivered
- Activities, schedules, outputs and planned objectives would be identified and co-development together with the research managers and principal investigators of the project

Learning and Knowledge Management

- Capturing the findings from the projects, institutionalising learning and organizing the information produced
- Organizing reflection workshops and learning reviews: critically assess results, identify success factors, and analyse challenges
- Outcomes: learning briefs, success stories, case studies and technical notes
- Embedding these practices within institutional processes, learning and knowledge management strengths organisational memory, avoids duplication of efforts, and fosters a culture of continuous improvement

Impact Assessment

- Evidence-based analysis of how ICAR's research, technology, policies and capacity building initiatives could contribute to increasing agricultural productivity, sustainability and livelihoods
- Organizing reflection workshops and learning reviews: critically assess results, identify success factors, and analyse challenges
- Measure benefits across economic, social, environmental and policy dimensions to estimate returns on investment of ICAR research, and inform strategic planning and resource allocation
- Embedding these practices within institutional processes, learning and knowledge management strengths organisational memory, avoids duplication of efforts, and fosters a culture of continuous improvement

Generalized Framework

Theory of Change

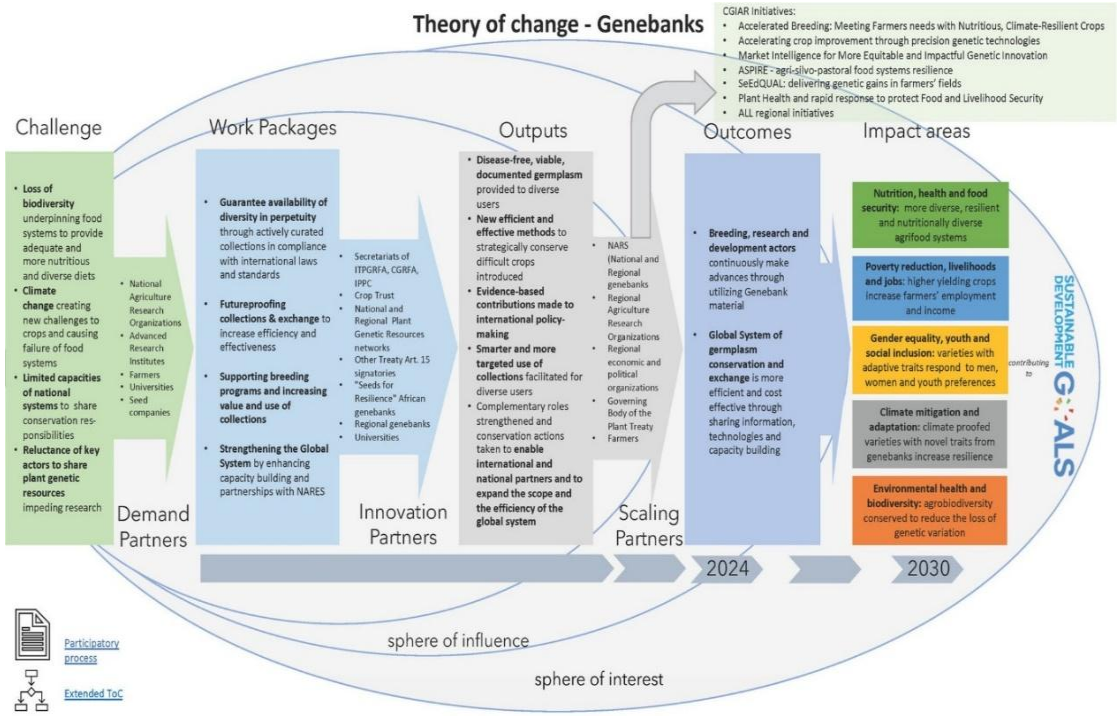


Figure 1. Theory of change for the Gene bank initiative

1) Defining the long-term impact
2) Identifying outcomes
3) Mapping outputs
4) List activities/inputs
5) Identifying assumptions
6) Defining indicators
7) Validating the pathways
8) Visualise the ToC

Generalized Framework

Impact Pathway

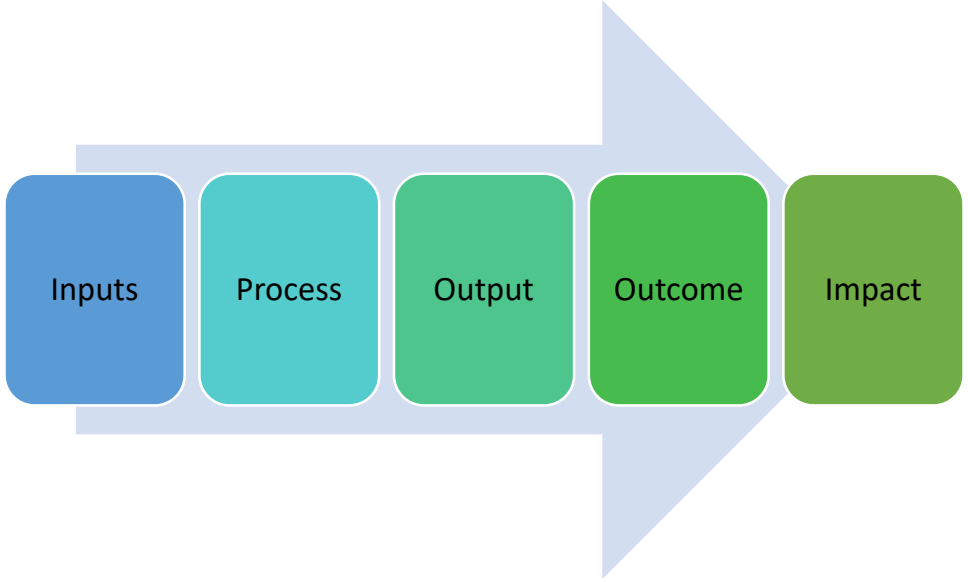


Figure 2. A representative impact pathway

- Impact pathway is a focused, operational component of a ToC
- **purpose** is to make explicit chain of events that lead from research activities to impact
- **scope** is narrower than the ToC as it focuses on translating the inputs into impact rather than the broader learning process

Framework Example: Development of Heat Tolerant Wheat Variety – Theory of Change

Problem: Rising temperatures (heat stress during grain-filling) reduce wheat yields by 10–25% in India's Indo-Gangetic Plains. Existing varieties lack adequate heat tolerance. ICAR research investment aims to develop climate-resilient varieties for sustained food security.

INPUTS	PROCESS	OUTPUTS	OUTCOMES	IMPACT
• ICAR project funding • Wheat germplasm (500+) • Research scientists (FTEs) • Laboratories & field sites	• Phenotyping & genotyping • Crossing & selection cycles • Multi-location field trials • Farmer participatory trials • CVRC regulatory process	• 5+ elite heat-tolerant lines • CVRC-released variety (≥1) • 10+ research publications • 50 q breeder seed • Training modules (≥3)	• 30%+ farmer adoption (Yr 5) • 2 lakh ha area under variety • +15% yield vs check • 60%+ extension staff trained • Reduced crop failure risk	• National wheat +0.5 Mt • Farmer income +₹3,500/ha • IRR ≥15% • SDG 2 & NDC contribution • Climate-resilient wheat system

2026 → 2027 → 2028 → 2029–30

Key Performance Indicators: Inputs

These KPIs track whether sufficient resources are mobilised and research activities are implemented as planned (Years 1–3).

KPI	Definition / Measurement	Target	Frequency
Inputs			
Annual research budget utilised (%)	% of approved project budget disbursed and spent on schedule	≥90%	Annual
Germplasm accessions screened (no.)	Wheat accessions evaluated for heat tolerance in glasshouse & field	≥500	Year 1–2
Scientific staff deployed (FTEs)	Full-time equivalents assigned to project activities	≥8 FTEs	Annual
Lab & field infrastructure readiness (%)	% of required facilities (labs, trial sites) fully operational	100%	Year 1
Germplasm data entered in system (%)	% screened accessions with complete records in NBPGR database	≥95%	Annual

Key Performance Indicators: Process

These KPIs track whether sufficient resources are mobilised and research activities are implemented as planned (Years 1–3).

KPI	Definition / Measurement	Target	Frequency
PROCESS			
Field trials conducted (no.)	Multi-location trials completed per cropping season	≥12/yr	Annual
Crossing combinations made (no.)	Unique crosses attempted in breeding nursery	≥200	Year 1–3
Marker-assisted selection cycles (no.)	Genomic screening cycles completed against heat-tolerance QTLs	≥3	Year 2–4
Farmer participatory variety trials (no.)	On-farm PVTs with real farmers per season	≥30	Year 3–4
Regulatory dossier submitted to CVRC	Go/No-Go: CVRC application filed (Yes/No + date)	Yes (Yr 4)	Once
Stakeholder consultations held (no.)	Workshops with farmers, traders, millers & SAUs	≥4/yr	Annual

Key Performance Indicators: Outputs

Output KPIs measure tangible deliverables from research; Outcome KPIs capture behavioural and system change at farmer and extension level (Years 3–6).

KPI	Definition / Measurement	Target	Frequency
OUTPUTS			
Peer-reviewed publications (no.)	ISI/Scopus papers published from project outputs	≥10	Annual
Heat-tolerant elite lines developed (no.)	Lines showing ≥10% yield advantage over check under heat stress	≥5	Year 3
Variety released by CVRC (no.)	New notified variety registered under Seeds Act 1966	≥1	Year 4–5
Breeder seed produced (quintals)	Certified breeder seed available for class seed multiplication	≥50 q	Year 4
FLD training modules developed (no.)	Extension/Front Line Demo packages developed and validated	≥3	Year 3
Dissemination materials published (no.)	Leaflets, videos, advisory notes in regional languages	≥5	Year 4

Key Performance Indicators: Outcomes

Output KPIs measure tangible deliverables from research; Outcome KPIs capture behavioural and system change at farmer and extension level (Years 3–6).

KPI	Definition / Measurement	Target	Frequency
OUTCOMES			
Farmer adoption rate (%)	% of target farmers growing new variety within 3 yrs of release	≥30%	Annual (Yr 5+)
Area under new variety (lakh ha)	Total cultivated area sown with the released heat-tolerant variety	≥2 lakh ha	Year 5–7
Yield under heat stress (t/ha)	Average yield of new variety vs check variety in heat-stress season	+15% vs check	Annual
Reduction in crop loss (%)	Reduction in heat-stress-induced yield loss vs. traditional variety	<10% loss	Year 5
Extension staff trained (%)	Proportion of state/district extension officers trained on new variety	≥60%	Year 4
Farmer awareness of variety (%)	Survey: % farmers aware of heat-tolerant variety in target districts	≥50%	Year 4

Key Performance Indicators: Impact & Underlying Assumptions

IMPACT KPIs (measured 5–10 years post-release via independent impact assessment studies)

KPI	Definition / Measurement	Target	Frequency
Incremental wheat yield (Mt, national)	Change in national wheat production attributable to adoption (econometric model)	+0.5 Mt by 2032	Every 3 yrs
Economic rate of return (IRR %)	Benefit-cost analysis: net benefits relative to total research investment	IRR ≥15%	End of project
Farmers' net income gain (₹/ha)	Average income gain from adoption of new variety vs baseline	+₹3,500/ha	Year 5 & 7
% reduction in heat-shock vulnerability	% of adopting farmers reporting reduced crop failure incidence	≥25% reduction	Year 7
Additional persons fed (million)	Incremental wheat supply converted to person-equivalents fed	+5 million	Year 8–10
SDG 2 / NDC contribution score	Rating by independent evaluator: contribution to zero hunger & climate goals	High/Moderate	Mid-term, End-term

KEY ASSUMPTIONS FOR THE THEORY OF CHANGE TO HOLD

Scientific	Adequate genetic diversity in germplasm for heat tolerance; trait is heritable and stable across Indo-Gangetic environments
Institutional	CVRC approves variety on merit within 4 years; sustained ICAR budget allocation throughout project duration
Extension	State agriculture departments and KVKs effectively communicate and demonstrate the variety to target farmers
Market	Seed companies/NSC take up multiplication; new variety fetches comparable market price to existing varieties
Climate	Heat stress frequency continues to increase as projected, maintaining farmer demand for heat-tolerant varieties

From Monitoring to Impact: A Linear Process

STAGE 1

Monitoring & Evaluation

KPIs tracked over time

STEP 1

Monitor with KPIs

Project-level Key Performance Indicators are captured and tracked continuously, giving real-time visibility into progress.

STAGE 2

Learning

Insights linked to project

STEP 2

Learn from the data

Patterns, gaps, and successes from M&E data are surfaced and linked back to the project — turning observation into actionable insight.

STAGE 3

Impact Assessment

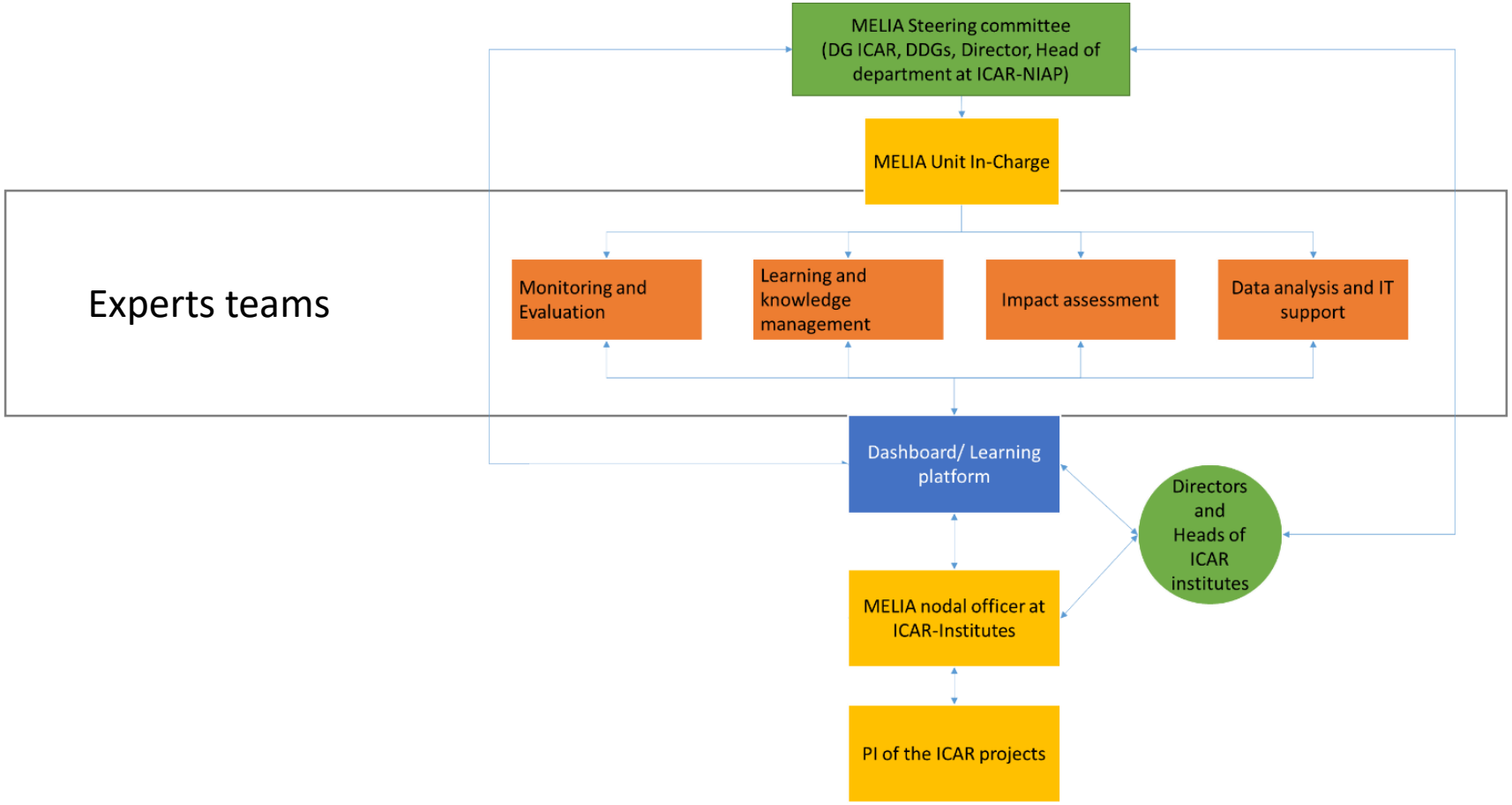
Internal or external review

STEP 3

Assess the impact

A formal review (internal or external) evaluates the project's outcomes against its goals and documents the demonstrated impact.

Structure of MELIA Unit



Approaches to KPI Data Collection

Comparing manual submission vs. automated extraction from RPPs

APPROACH 1 · Manual Submission

Process Flow

Researcher enters KPIs (Excel) → Nodal officer verifies → Director verifies → Submitted to M&E Unit

TIME PER SUBMISSION
High (Excel filling)

RESOURCE NEED
Low setup cost

Key Characteristics

- Time-intensive; repetition is expensive
- Faster initial rollout (just send Excel sheets)
- Compliance dependent — repeated follow-ups likely
- Manual entry → higher risk of errors & inconsistency

APPROACH 2 · Automated Extraction

Process Flow

Researcher uploads RPP (PDF/Word) → System auto-extracts KPIs → Nodal officer verifies → Director verifies

TIME PER SUBMISSION
3–5 min upload

RESOURCE NEED
High (system funding)

Key Characteristics

- Less time-intensive; runs at regular intervals
- Longer development/setup time upfront
- Higher likelihood of compliance (built into workflow)
- Auto-verified data → consistent and audit-ready

Trade-off: Manual = *lower setup, higher recurring effort, weaker compliance* | Automated = *higher setup, lower recurring effort, stronger compliance*

Information Flow

Role-based pathway from project reporting to ICAR-level decision-making

STEP 01

Project Principal Investigator

Annual reporting

Submits project KPIs, outputs, outcomes, and learnings via the MELIA dashboard or RPP upload.

STEP 02

Institute MELIA Nodal Officer / Director

Validation & portfolio view

Verifies data quality, aggregates institute-level performance, and identifies thematic learning needs.

STEP 03

MELIA Unit (ICAR-NIAP)

Dashboard & tracking indicators

Synthesises results across institutes, runs analytics, prepares evidence-based reports and meta-analyses.

STEP 04

Steering Committee & ICAR Management

Adaptive management & strategy

Uses synthesised evidence for policy inputs, resource allocation, and future programme design.

Bottom-up evidence feeds top-down decisions — *every level adds value before passing it on.*

Governance Structure & Roles

Who decides what, and how the MELIA system holds itself accountable

MELIA STEERING COMMITTEE

Composition DG ICAR (Chair) · SMD leads · Director NIAP · HoD Markets & Trade · HoD Technology & Sustainability · One independent MELIA expert

Approves Annual MELIA plans · Major evaluation ToRs · Annual Results & Impact Report

MELIA Unit

(at ICAR-NIAP)

- Standard-setting & quality assurance
- Commissioning evaluations
- Data governance
- Synthesis & meta-analysis
- Knowledge products & capacity building
- Helpdesk for projects & institutes

Nodal Officer

(at each ICAR Institute)

- Portfolio oversight
- Quarterly performance & learning reviews
- Identifying thematic evaluations
- Synthesis at institute level
- First line of data validation

Project PIs

(Principal Investigators)

- Develop & maintain MELIA plans
- Timely data collection & reporting
- Document use, uptake, and adoption
- Participate in reflection workshops

Institute Directors

(ICAR Institutes)

- Maintenance of institute dashboards
- Data quality checks & sign-off
- Compliance with data governance
- Hosting learning events

Designed for minimum reporting burden — ***with built-in incentives*** to support active participation by nodal officers and PIs.

Expectations

What each stakeholder gains from the MELIA Unit

FOR THE COUNCIL

ICAR Leadership

- **Evidence-based inputs** — for strategic planning, budgeting, and policy decisions
- **Clear demonstration** — of returns and impact of ICAR investments
- **Standardised reporting** — credible performance and impact reporting across ICAR

FOR INSTITUTIONS

ICAR Institutes

- **Integrated monitoring** — with minimal additional reporting burden
- **Capacity building** — and technical support for M&E, learning, and impact assessment
- **Actionable insights** — at portfolio level for programme improvement

FOR SCIENTIFIC STAFF

Researchers & PIs

- **Clear guidance** — and handholding on indicators, data collection, and reporting
- **Constructive feedback** — and learning to improve project performance
- **Documentation support** — to demonstrate the impact of their research

Three audiences, one promise: MELIA delivers *actionable evidence* where it's needed without adding to the workload.

Timelines

Roadmap for setting up the MELIA Unit

Kick-off
Month 0

Workshop / Seminar with CGIAR MELIA units

MELIA NIAP team aligns with global best practices through a launch workshop with CGIAR MELIA units.

3 months
Q1

Learning & KM Manual

Manual prepared and shared across the unit covering reflection workshops, knowledge products, and learning reviews.

6–7 months
Q2

Impact Assessment Manual

Methods, design selection guidance, and case-study formats — completed in another 3–4 months after Q1.

6 months
(parallel)

M&E Indicator Manual + KPI co-development with SMDs

Indicator definitions, baselines, and targets developed jointly with SMDs — providing the structure for the dashboard.

1 year
Year-end

Dashboard Manual finalised

Dashboard architecture, data flows, and user manuals fully documented and operational across institutes.

Within 12 months: all four manuals ready, KPIs co-developed with SMDs, and the MELIA dashboard operational.

Thank you